LOW-LEVEL DESIGN (LLD)

1. Module-Level Design

- Document Processing Module → Loads PDF
- Chunking Module → Splits text
- Embedding Module → Converts text to vectors
- Vector Storage Module → Stores embeddings
- Retrieval Module → Fetches relevant chunks
- Query Processing Module → Handles user queries
- Graph Execution Module → Controls flow
- HITL Module → Handles escalation

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2. Data Structures

Document

- page_content: string
- metadata: dictionary

Chunk

- text: string
- id: integer

Embedding

- vector: float array

State (LangGraph)

- query: string

- response: string

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3. Workflow Design (LangGraph)

Nodes:

- Processing Node

Edges:

- Input → Process → Output

State:

- Carries query and response

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4. Conditional Routing Logic

Escalation triggered when:

- Response length < 30
- Contains “don’t know”
- No relevant context
- Complex query

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5. HITL Design

- System detects low-quality answer
- Escalates to human agent
- Returns fallback response

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6. API Design

Input:

```
{
"query": "string"
}
```

Output:

```
{
"response": "string"
}
```

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7. Error Handling

- Missing PDF → error message
- No chunks found → fallback
- LLM failure → retry or escalate